2

Electron–phonon interactions

Electron–phonon interactions describe how the energy of an electronic charge is affected by alterations of the positions of the atoms of the encompassing medium. Polaron problems usually take a charge carrier's energy to depend linearly on shifts of nuclear positions. Here, for simplicity, the medium is represented as a continuum. Then the net effect of deforming the continuum by $\Delta(\boldsymbol{u})$ at positions designated by $\boldsymbol{u}$ is to shift the effective potential energy of a carrier at $\boldsymbol{r}$ by

$$V(\boldsymbol{r}) = \int d\boldsymbol{u} Z(\boldsymbol{r}-\boldsymbol{u})\Delta(\boldsymbol{u}), \tag{2.1}$$

where $Z(\boldsymbol{r}-\boldsymbol{u})$ defines the magnitude and range of the electron–phonon interaction.

2.1 Long-range electron–phonon interaction

Ionic and polar materials support a long-range component of the electron–phonon interaction. Pairing anions with cations enables the Coulomb potential of these materials to be represented as that from an array of electric dipoles. Altering the separation between the ions of a dipole shifts the potential seen by a carrier. The modulation of the carrier's potential energy at $\boldsymbol{r}$ due to altering the charge separation of a dipole centered at $\boldsymbol{u}$ is governed by

$$Z_{\mathrm{LR}}(\boldsymbol{r}-\boldsymbol{u}) \equiv -\frac{\sqrt{\dfrac{e^2}{4\pi}\left(\dfrac{1}{\varepsilon_\infty}-\dfrac{1}{\varepsilon_0}\right)\dfrac{k}{V_{\mathrm{c}}}}}{|\boldsymbol{r}-\boldsymbol{u}|^2}\cos\theta, \tag{2.2}$$

where (as illustrated in Fig. 2.1) θ is the angle between the electric dipole and the vector $\boldsymbol{r}-\boldsymbol{u}$. The constants in Eq. (2.2) are the carrier's charge e, the material's

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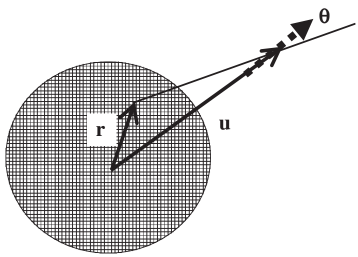

Fig. 2.1 An electric dipole (dotted arrow) directed along the vector from the carrier's centroid $\boldsymbol{u}$ interacts with its charge (hatched area) at $\boldsymbol{r}$.

optical and static dielectric constants ε_∞ and ε_0, the Hooke's law stiffness constant for the electric dipole's charge separation k, and the unit cell volume V_c. This combination of physical parameters relates a modulated electric dipole to the polarization-adjusted electric potential that it produces.

A material's dc polarization is proportional to $1 - 1/\varepsilon_0$. This polarization contains contributions from displacing electrons as well as from displacing relatively massive and slow-moving ions. By comparison the optical polarization $1 - 1/\varepsilon_\infty$ only contains contributions from electrons since only they can move fast enough to follow a high-frequency oscillating electric field. Thus, $1/\varepsilon_\infty - 1/\varepsilon_0$ measures the slow polarization arising from displacing ions. This difference of the reciprocal of the dielectric constants tends to be small for covalent materials since there $\varepsilon_0 \approx \varepsilon_\infty$. Hence the associated long-range electron–phonon coupling also tends to be small for covalent materials. By contrast, the long-range electron–phonon coupling is of major importance in ionic crystals such as alkali halides when $\varepsilon_0 \approx 2\varepsilon_\infty$. The long-range electron–phonon coupling is exceptionally strong in ferroelectric and extremely polar materials as they are characterized by $\varepsilon_0 \gg \varepsilon_\infty$.

The potential well produced by ions' relaxation about a static point charge is a useful fiducial situation when discussing self-trapping. Ions assuming shifted equilibrium positions in response to the presence of the static point charge produce a Coulomb-like well

$$V_{\mathrm{eq}}^{\mathrm{LR}}(\boldsymbol{r}) = -\left(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0}\right)\frac{e^2}{|\boldsymbol{r}|}. \tag{2.3}$$

The magnitude of this Coulombic potential measures the strength of the long-range electron–phonon interaction. The potential well generated by the long-range electron–phonon interaction about a confined carrier of radius R is schematically illustrated in Fig. 2.2. As discussed in Section 4.3, the well depth is proportional to $1/R$.

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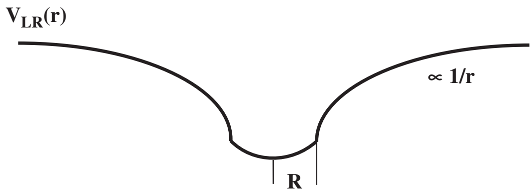

Fig. 2.2 A potential well forms about a confined carrier of radius R as its long-range electron–phonon interaction induces shifts of the equilibrium positions of surrounding ions.

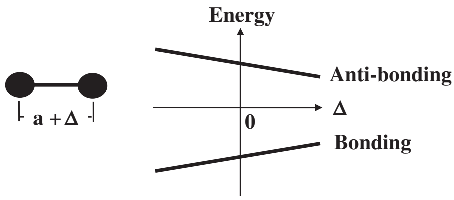

Fig. 2.3 The energies of the bonding and anti-bonding states of a covalent bond change as its length shifts from its carrier-free equilibrium value a by Δ.

2.2 Short-range electron–phonon interactions

The short-range electron–phonon interaction results when the energy of a carrier depends on the strain where it resides. Within the continuum treatment the short-range electron–phonon interaction is represented by the contact interaction

$$Z_{\mathrm{SR}}(\boldsymbol{r}-\boldsymbol{u}) \equiv -F\delta(\boldsymbol{r}-\boldsymbol{u}), \tag{2.4}$$

where the constant F has the dimensions of a force. Strain-induced shifts of electronic transfer energies and the energies of bonding and anti-bonding states contribute to the short-range electron–phonon interaction. For example, as indicated in Fig. 2.3, compressing a covalent bond tends to lower the energy of the bonding state while increasing the energy of the anti-bonding state.

The lowering of the effective potential for a carrier on a single isolated structural unit (e.g. a bond) is used to reference the strength of the short-range interaction. Within the continuum treatment the carrier-induced strain is that which minimizes the sum of the carrier's energy, $-F\Delta(\boldsymbol{r})$, plus the Hooke's-law strain energy for the associated strain, $k\Delta(\boldsymbol{r})^2/2$, where k denotes the continuum's stiffness constant. The equilibrium strain is thus $\Delta_{\mathrm{eq}} = F/k$. The lowering of the potential energy of Eq. (2.1) when a carrier is confined to a single bond is

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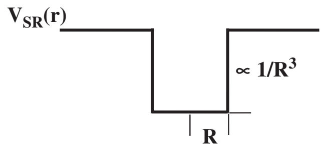

Fig. 2.4 A potential well forms about a confined carrier of radius R as its short-range electron–phonon interaction induces shifts of the equilibrium positions of atoms it contacts.

$$V_{\mathrm{eq}}^{\mathrm{SR}} = -\frac{F^2}{k}. \tag{2.5}$$

The potential well generated by the short-range electron–phonon interaction about a confined carrier of radius R is schematically illustrated in Fig. 2.4. As discussed in Section 4.3, the well depth is proportional to $1/R^3$.

The depth of this potential is typically less than 1 eV since values of F are usually 2–3 eV/Å with $k > 10$ eV/Å^2 for a covalent bond. By comparison, the depth of the potential produced by the long-range interaction in an ionic material, Eq. (2.3), is about 3 eV with $\varepsilon_0 \approx 2\varepsilon_\infty = 5$ and $r \approx 1$ Å. Since ionic and polar materials possess relatively strong electron–phonon interactions their effects have become known as *polaron* effects. Nonetheless, polaron effects arising from short-range electron–phonon interactions can be significant in ionic materials as well as in covalent solids.